Data Visualisation

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1. About the dataset

The dataset is about the world's billionaires in 2023 and is obtained from Kaggle.com.
It contains the billionaires' demographic information, industries and businesses. Also included in the dataset are several selected macroeconomic indices such as the CPI, GDP, gross teritary education enrollment, gross primary education enrollment, life expectancy, tax revenue, tax rate and population. All these indices refer to the country, in which the billionaire is living in.
This document only focuses on the data visualization and key findings.

2. Distribution

2.1 By Industry

```r
factor_category <- factor(df_Billionaires$category)
category1 <- as.data.frame(table(factor_category))

pl_cat <- ggplot(category1, aes(x = reorder(factor_category, -Freq),
                                y = Freq)) +
  geom_bar(stat = 'identity', aes(fill = Freq)) +
  scale_fill_viridis() +     #from package "viridis"
  geom_text(aes(label= Freq), vjust=-0.3, size=2.5) +
  theme_bw() +
  labs(color = NULL,
       y = "Frequency",
       x = "Industries") +
  theme(strip.background = element_blank(),
        axis.text.x = element_text(angle = 70,
                                   hjust = 1),
        legend.key.height= unit(0.3, 'cm'),
        legend.key.width= unit(0.3, 'cm'))
pl_cat
```

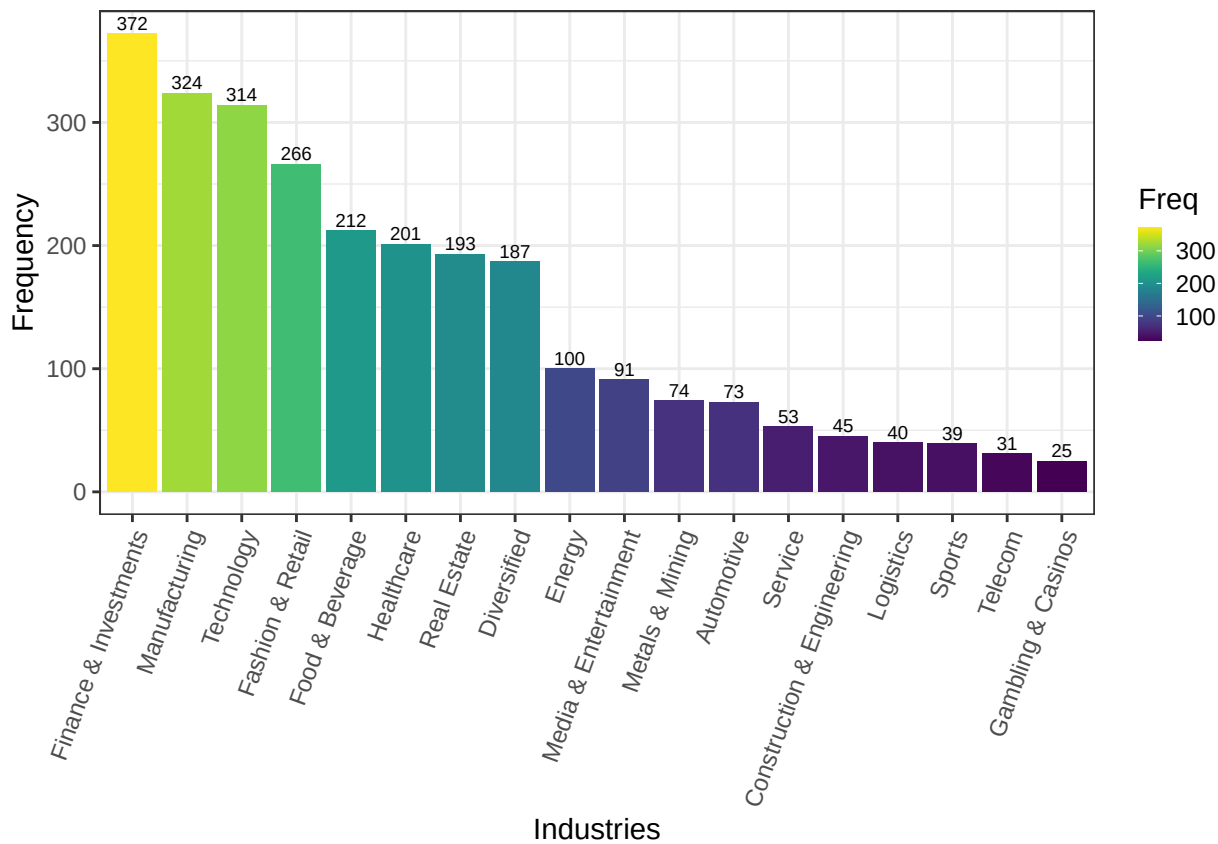

Figure 1: Distribution of world's billionaires by industry

The Finance & Investements industry has the highest number of billionaires (372). Other most populous sectors among the billionaires list are Manufacturing (372), Technology (314), Fashion & Retail (266) and Food & Beverage (212).

2.2 By Gender

```
factor_ges <- factor(df_Billionaires$gender)
ges <- as.data.frame(table(factor_ges)) %>%
  rename(Anzahl = Freq) %>%
  mutate(Anteil = Anzahl/2640) %>%
  mutate(Anteilp = scales::percent(Anteil, accuracy = 0.01))

plot_ges <- ggplot(ges, aes(x = "", y = Anteilp, fill = factor_ges)) +
  geom_bar(stat = 'identity', width = 0.6) +
  geom_text(aes(label = Anteilp),
            position = position_stack(vjust = 0.5), size= 4) +
  theme_void() +
  guides(fill = guide_legend(title = "Gender"))  +
  coord_polar("y", start=0) +
  labs(color = NULL)
plot_ges
```

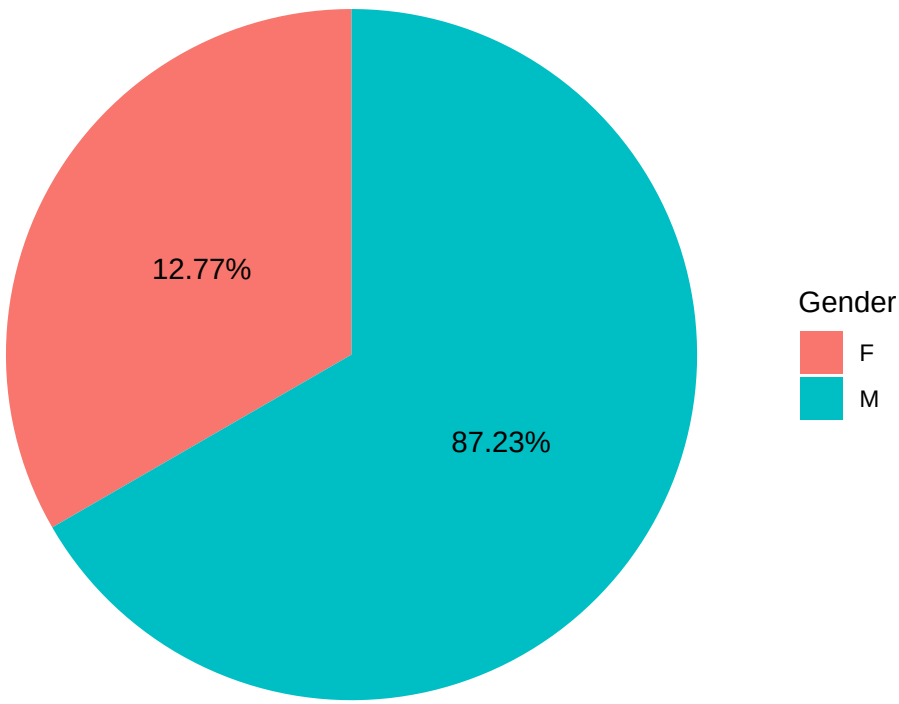

Figure 2: Distribution of world's billionaires by gender.

The majority of the billionaires (87.23%) are male, whereas only 12.77% are female.

2.3 By Country

```
factor_country <- factor(df_Billionaires$country)
country <- as.data.frame(table(factor_country))
country_sorted <- country[order(country$Freq),] %>%
  tail( n= 10L)

cols1 <- c( "pink",
            "#BF87B3","#BF87B3","#BF87B3","#BF87B3","#BF87B3",
            "#8255a1",
            "#8255a1",
            rgb(72/255, 32/255, 143/255),
            rgb(72/255, 32/255, 143/255))

plot_country <- ggplot(country_sorted, aes(x = reorder(factor_country, -Freq,
                                                       decreasing = FALSE),
                                           y = Freq)) +
  geom_bar(stat = 'identity', fill = cols1) +
  geom_text(aes(label= Freq), vjust=-0.3, size=3) +
  theme_bw() +
  labs(color = NULL,
       y = "Number of Billionaires",
       x = "Country") +
 theme(strip.background = element_blank(),
        axis.text.x = element_text(angle = 45,
                                   hjust = 1, size = 9),
       axis.text.y = element_text(size = 9))
plot_country
```

The United States has the most billionaires, followed by China, India and Germany.

2.4 By Status

```
factor_status <- factor(df_Billionaires$selfMade)
status <- as.data.frame(table(factor_status)) %>%
  mutate(Anteil = Freq/2640) %>%
  mutate(Anteilp = scales::percent(Anteil, accuracy = 0.01))

plot_status <- ggplot(status, aes(x = "", y = Anteilp, fill = factor_status)) +
  geom_bar(stat = 'identity') +
  theme_void() +
  geom_text(aes(label = Anteilp),
            position = position_stack(vjust = 0.5), size= 4.5) +
  guides(fill = guide_legend(title = "Status")) +
  scale_fill_manual(labels = c("inherited", "self-made"),
                    values=c("beige", "coral1")) +
  coord_polar("y", start=0) +
  labs(color = NULL)
plot_status
```

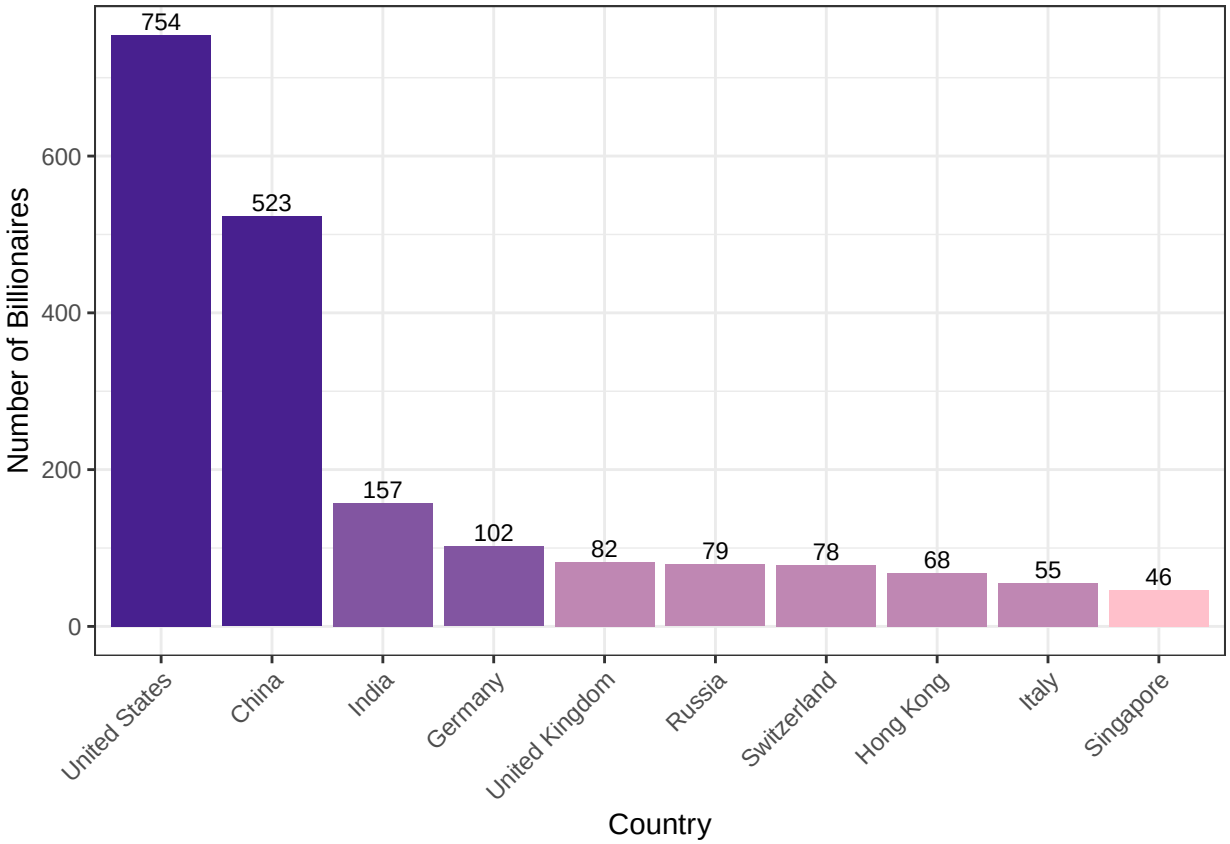

Figure 3: Distribution of world's billionaires by country

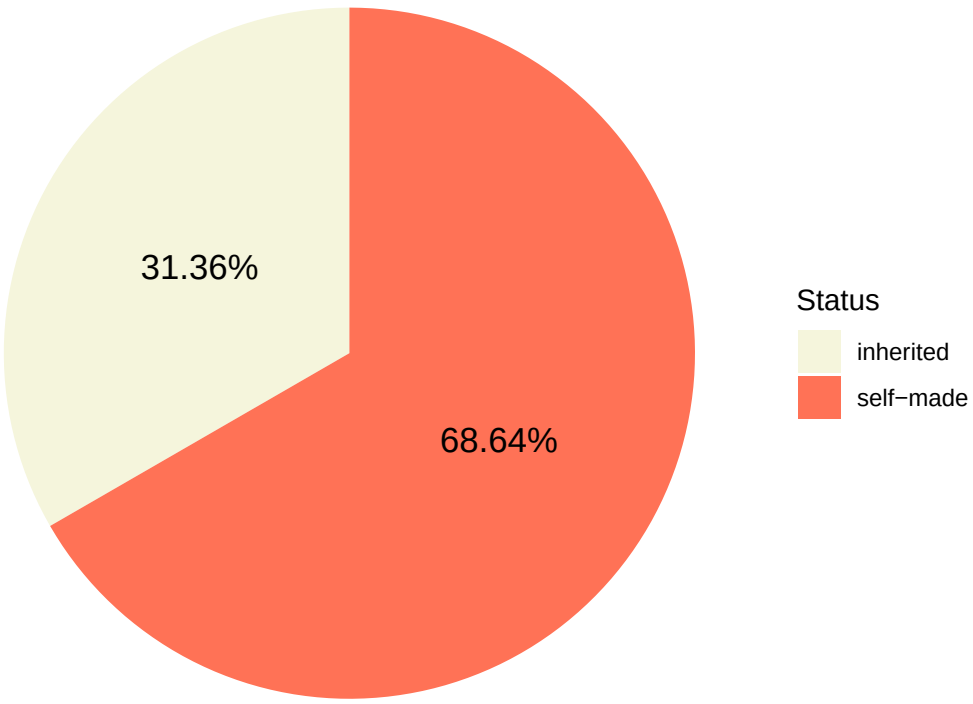

Figure 4: Distribution of word's billionaires by status

The largest part of the billionaires are self-made.

2.5 By Age

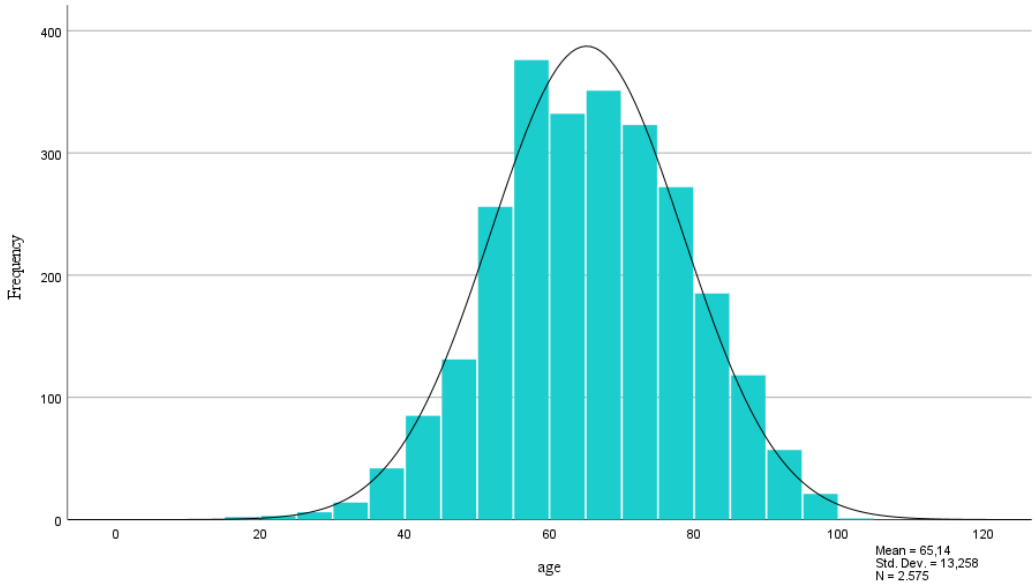

Figure 5: Age distribution of world's billionaires

Based on the normal curve, the population's age is approx. normally distributed, since the most common age falls between 55 and 60, but not at the peak of the normal curve (between 60-65 or 70-75).

2.6 Top 10 richest

```
top10 <- df_Billionaires %>%
  head(n = 10L)

plot_top10 <- ggplot(top10, aes(x = reorder(personName, finalWorth_bln),
                                y = finalWorth_bln, fill = category)) +
  geom_bar(stat = 'identity') +
  geom_text(aes(label = finalWorth_bln),hjust = -0.5, size = 1.6)  +
  guides(fill = guide_legend(title = "Category")) +
  theme_bw() +
  coord_flip() +
  labs(color = NULL,
       y = "Net wealth in billion USD",
       x = NULL,
       title = "Top 10") +
  theme(strip.background = element_blank(),
        axis.text.x = element_text(size = 8),
        axis.text.y = element_text(size = 6),
        legend.key.height= unit(0.3, 'cm'),
```

```
        legend.key.width= unit(0.3, 'cm'),
        legend.text = element_text(size = 6))
plot_top10
```

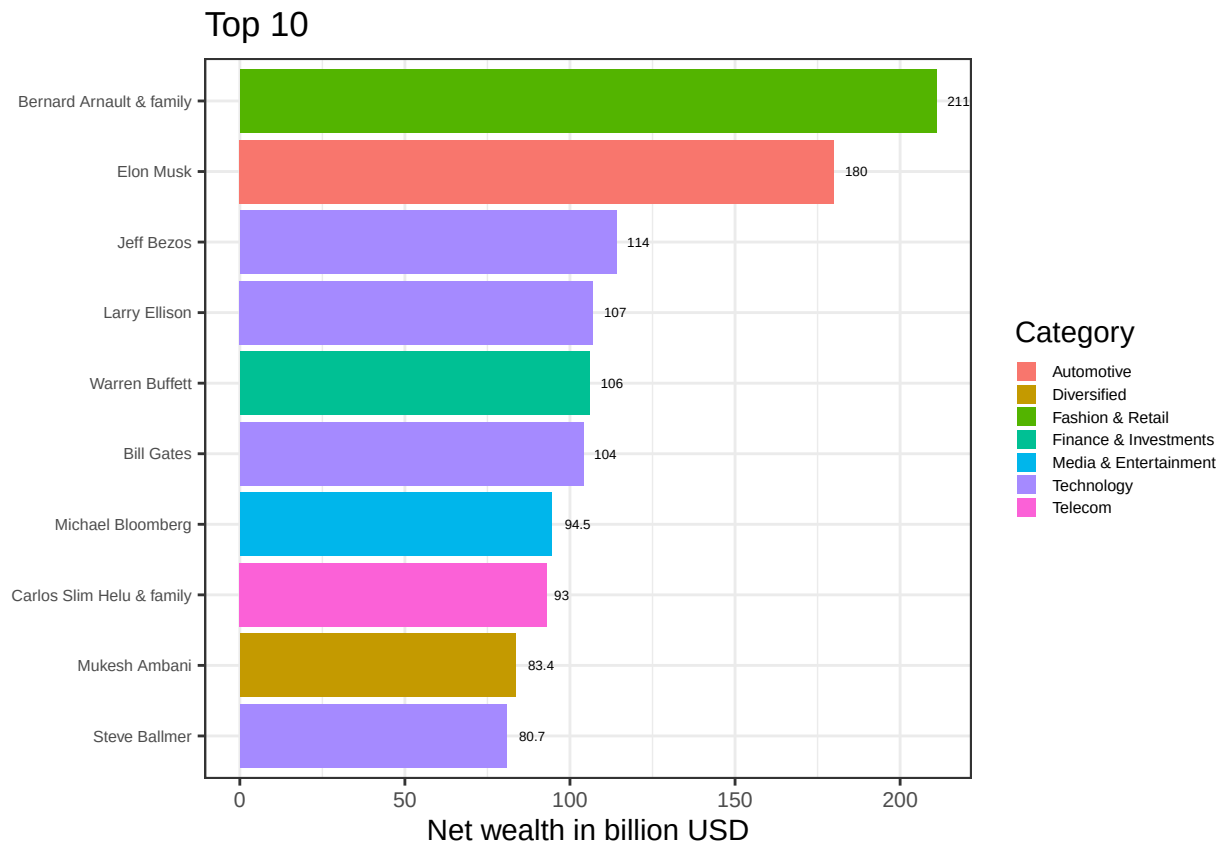

Figure 6: Top 10 richest people in the world (2023)

As of 4.4.2023, Bernard Arnault & family from the fashion & retail industry rank first among the world's 10 wealthiest people with the highest net worth of $211 billion.
Elon Musk is the world's second wealthiest person whose net worth stems from the automotive industry.
Technology is most populous sector among the world's top 10 since it has 4 billionaires named in the list, including Jeff Bezos ($114 billion), Larry Ellison ($107 billion), Bill Gates ($104 billion) and Steve Ballmer ($80.7 billion).

3 Crosstabulations

3.1 Industries and Age groups

```
age_groups <- ifelse(df_Billionaires$age < 35, "under 35",
                     ifelse(df_Billionaires$age < 55, "35 to under 55",
                            ifelse(df_Billionaires$age < 75, "55 to under 75",
                                   "over 75")))
```

```
df_Billionaires$agegroups <- age_groups
df_agegroups_industries <- as.data.frame(xtabs(~df_Billionaires$industries +
                                                df_Billionaires$agegroups))


v1 <- df_agegroups_industries$Freq[1:18] / sum(df_agegroups_industries$Freq[1:18])
v2 <- df_agegroups_industries$Freq[19:36] / sum(df_agegroups_industries$Freq[19:36])
v3 <- df_agegroups_industries$Freq[37:54] / sum(df_agegroups_industries$Freq[37:54])
v4 <- df_agegroups_industries$Freq[55:72] / sum(df_agegroups_industries$Freq[55:72])

v5 <- c(v1,v2,v3,v4)
df_agegroups_industries$v5 <- v5

df_ageindus_final <- df_agegroups_industries %>%
  mutate(v5= scales::percent(v5, accuracy = 0.01))

plot_treemap <- df_ageindus_final %>%
  ggplot(aes(area = Freq, fill = df_Billionaires.industries,
             group = df_Billionaires.agegroups, subgroup = df_Billionaires.industries,
             label = paste0(df_Billionaires.industries, "\n",v5))) +
  geom_treemap() +
  facet_wrap(~factor(df_Billionaires.agegroups, c("under 35", "35 to under 55",
                                                  "55 to under 75", "over 75")))  +
  theme(strip.text = element_text(face = "bold", size = 10, color = "black"),
        legend.text = element_text(size = 6),
        legend.key.height= unit(0.3, 'cm'),
        legend.key.width= unit(0.3, 'cm')) +
  geom_treemap_text(grow = F, reflow = T, size = 7, colour = "black") +
  geom_treemap_subgroup_border(colour = "white") +
  scale_fill_tableau(palette = "Hue Circle") +
  guides(fill = guide_legend(title = "Industry"))
plot_treemap
```

It is to observed that the percentage of technology declines across the age groups (from 28%, 25.10%, 10.06% to 4.89%). Younger billionaires tend to be more involved in the tech industry than older.
The opposite applies for Investment and Finance, where the percentage rises as age increases (from 0%, 13.62%, 14.33% to 14.53%). Older billionaires are tendencially more active in the investment and finance industry than younger.
The observations suggest that there could be a correlation between age and industry. For this hypothesis, the chi-square tests were computed.

The p-value of 0.000 - displayed under "asymptotic significance (2-sided)" in row "Pearson Chi-Square"- is below the significance level under the assumption that the significance level is set at 0.05. The null hypothesis is therefore rejected and the observations were true. There is a statistically significant association between age groups and industries.

3.2 Net Worth and Industries

```
df_anova <- df_Billionaires %>%
  group_by(industries) %>%
  summarize(min = min(finalWorth_bln),
```

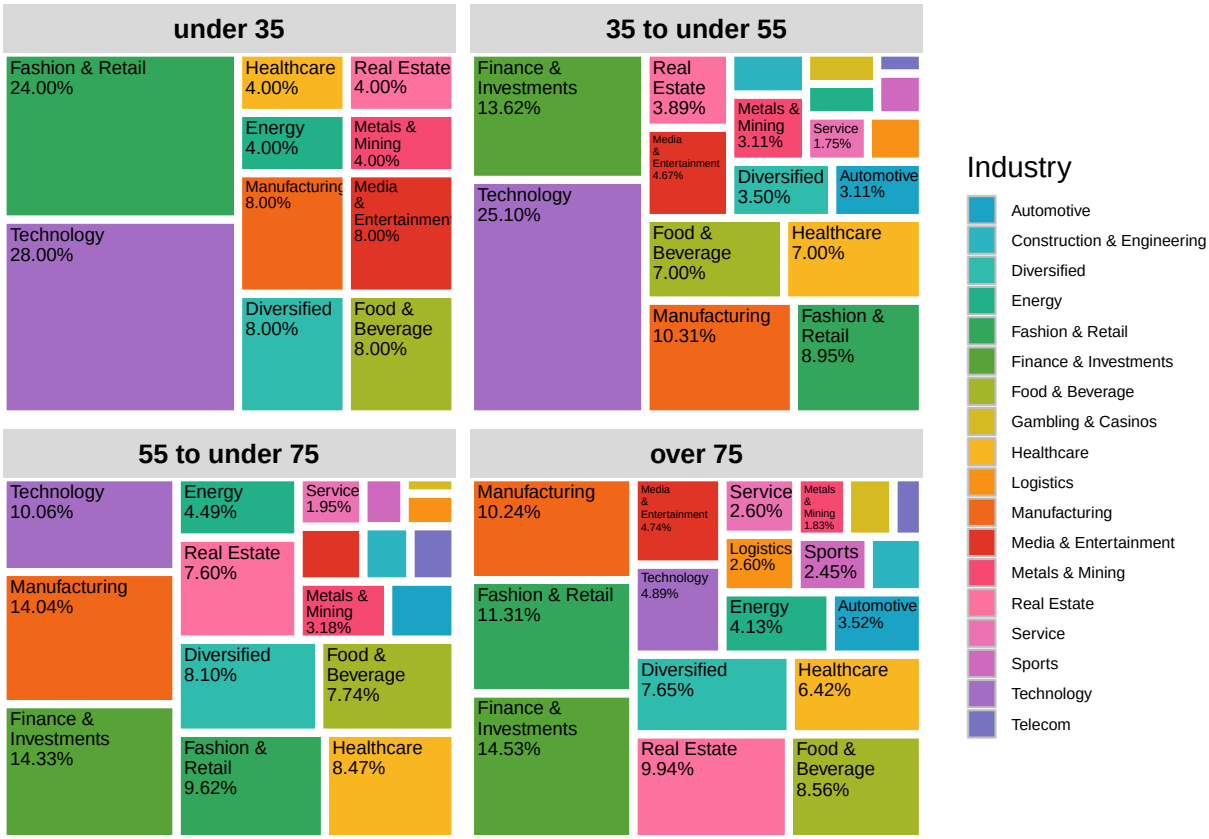

Figure 7: Percentage of different industries in different age groups

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	216,101[a]	51	,000
Likelihood Ratio	214,688	51	,000
N of Valid Cases	2575		

a. 19 cells (26,4%) have expected count less than 5. The minimum expected count is ,24.

Figure 8: Chi-Square Tests - computed in IBM SPSS

```
            max = max(finalWorth_bln),
            median = median(finalWorth_bln),
            mean = mean(finalWorth_bln))
df_anova
```

```
## # A tibble: 18 x 5
##    industries                    min   max median  mean
##    <chr>                       <dbl> <dbl>  <dbl> <dbl>
##  1 Automotive                    1   180     2.6   7.20
##  2 Construction & Engineering    1    13.7   2.1   2.63
##  3 Diversified                   1    83.4   2.3   4.84
##  4 Energy                        1    25.5   2.45  4.54
##  5 Fashion & Retail              1   211     2.5   6.39
##  6 Finance & Investments         1   106     2.6   4.31
##  7 Food & Beverage               1    68     2.5   4.52
##  8 Gambling & Casinos            1    35     2.8   4.82
##  9 Healthcare                    1    22.6   2.1   3.2
## 10 Logistics                     1    39.1   3     5.99
## 11 Manufacturing                 1    29.7   2     3.15
## 12 Media & Entertainment         1    94.5   2.5   4.70
## 13 Metals & Mining               1.1  27     2.6   6.04
## 14 Real Estate                   1    29.5   2.3   3.41
## 15 Service                       1    21.2   2.6   3.27
## 16 Sports                        1    13.3   2.1   3.45
## 17 Technology                    1   114     2.2   5.98
## 18 Telecom                       1    93     2.8   6.56
```

```
plot_anova <- ggplot(df_anova, aes(x = reorder(industries,-mean,
                                               decreasing = FALSE), y = mean)) +
  geom_bar(stat = "identity",aes(fill = mean)) +
  scale_fill_viridis(option = "magma", name = paste0("average", "\n","net worth"))+
  theme_bw() +
  geom_text(aes(label = scales::number(mean, accuracy = 0.01)),
            vjust = -0.3, size = 2.5) +
  labs(y = "average net worth $bn",
       x = "industry") +
  theme(strip.background = element_blank(),
        axis.text.x = element_text(angle = 45,
                                   hjust = 1))

plot_anova
```

Despite the findings from Figure 1 that finance & investments has the most billionaires, its average net worth only ranks 12th among 18 sectors. Reversely, the automotive sector, which places 12th in Figure 1, becomes the top sector with the highest average net worth of $7.20 USD.

```
anova1 <- aov(df_Billionaires$finalWorth_bln ~ df_Billionaires$industries)
summary(anova1)
```

```
##                               Df Sum Sq Mean Sq F value   Pr(>F)
## df_Billionaires$industries   17   4007  235.69    2.46 0.000764 ***
```

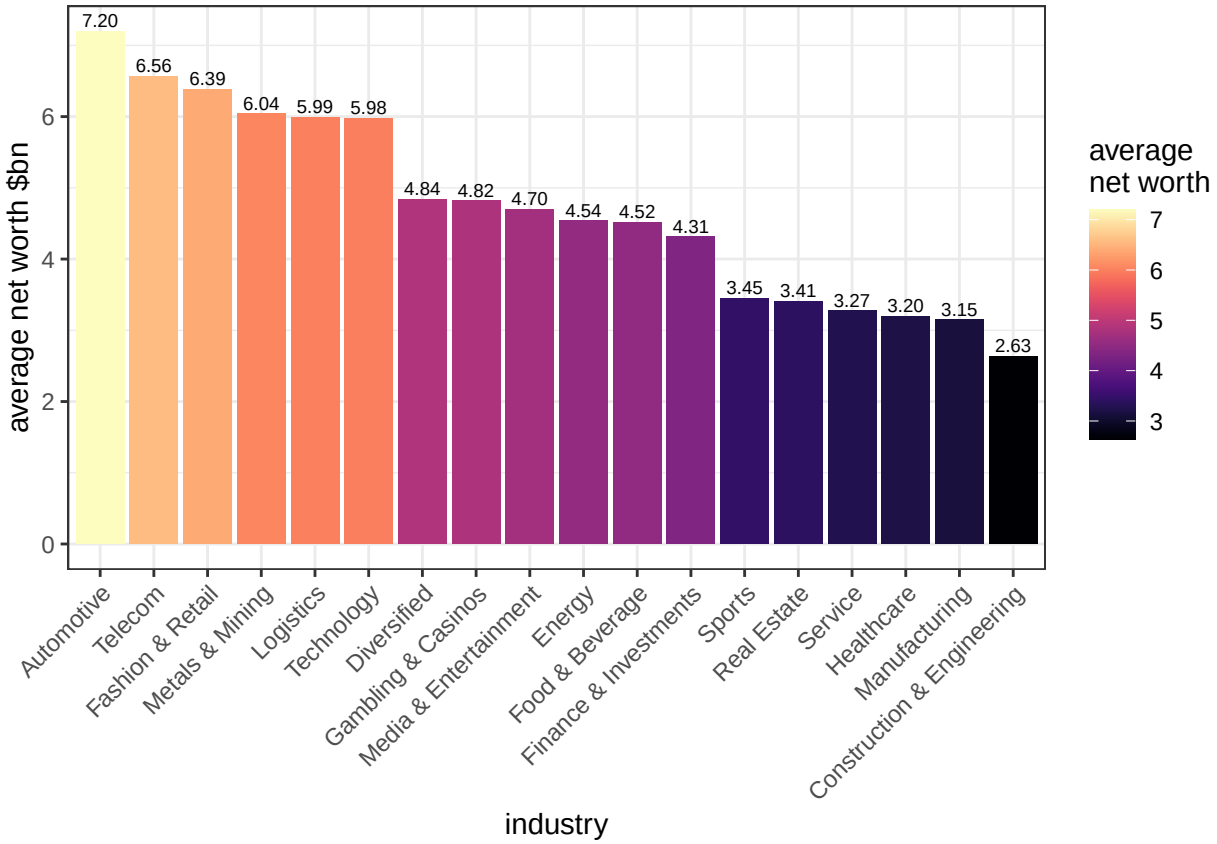

Figure 9: Average net worth by industry

```
## Residuals                  2622 251217   95.81
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The Analysis of Variance returns a p-value of 0.000764 (<0.05) showing that the relationship between average net worth and industry is statistical significant.